Crossing the Bifrost: Towards an open access FAIR HTR model for Old Norse manuscripts.

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Aim

This paper presents the initial results of a project devoted to creating the first-ever FAIR model for handwritten text recognition (HTR) of Old Norse-Icelandic manuscripts, accompanied by an open-access dataset of ground truths for manuscripts from the 13 th and 14 th centuries–the golden age of manuscript production in Iceland. Compared with other medieval languages written in Latin scripts, Old Norse-Icelandic does not benefit from the same amount of training data, which are crucial for developing strong HTR models. Therefore, we turn to an approach applied by scholars in other under-resourced fields, such as Arabic, to find sustainable solutions for the acquisition of texts. The goal of this paper is not only the presentation of another HTR model and its underlying dataset.[1] By incrementally evaluating the accuracy of the model, we provide insights into the amount of data actually needed to produce usable transcriptions. We provide reflections on the accessibility and influence of technology on research in smaller, often under-resourced disciplines, as well as on the position of individual researchers who are based outside large DH centres benefiting from state-of-the-art infrastructure. We also position our experiments with HTR in the broader context of research questions concerning Icelandic palaeography and scribal culture. One of the important points demonstrated by our paper is that all training and testing were done with Kraken (Kiessling 2019), run locally on a 2020 M1 MacBook Air (8 CPU, 16 GB). A comparable machine today can be purchased for around EUR 1.000, making it more accessible for many researchers than obtaining access to GPUs.

Background

While medieval Iceland is by many exclusively associated with vikings, volcanoes, and glaciers, few know that it was also home to impressive cultural production including the composition of native literature and the production of countless manuscripts. The Old Norse-Icelandic literary corpus has hitherto received little attention from the HTR community. While there are over 900 Icelandic manuscripts produced between the twelfth and the sixteenth centuries, only a fraction of them has been used in scholarly editions and even fewer in digital ones. A significant portion of these manuscripts (at least 50%), however, has already been digitised, and thousands of high-quality digital images are freely available online. Hence, there is great and hitherto unexploited potential for leveraging computer-assisted methods to approach this material and make it more accessible to the scholarly community by producing auto-generated transcriptions. The applications for auto-generated transcriptions are countless and already play an important role in digital palaeographical and philological workflows in certain fields of medieval studies, e.g. Latin, German, and French, but are much less common in smaller fields. Captivated by the great promise of generic HTR models that they can be easily and quickly fine-tuned for new contexts without great computational effort, we decided to test the applicability of existing HTR models for the Icelandic context and the amount of data needed to fine-tune them.

Dataset and Method

Starting with the Icelandic TranscriboQuest2024 dataset (Kapitan et al. 2024), we used eScriptorium for additional data acquisition (Stokes et al. 2021; Kiessling et al. 2019), creating the corpus summarised in Table 1. The transcriptions of all manuscripts, with the exception of AM 334 fol., the transcription of which is based on the digital facsimiles prepared with the aid of the printed edition (*Grágás* 1879), are based on the MENOTA corpus (Haugen et al. 2019). The layout for most of the manuscripts was corrected by Ariane Pinche in eScriptorium, and the others were done by Chahan Vidal-Gorène in Calfa Vision (Vidal-Gorène et al. 2021).

For the model, we took our point of departure in the recent efforts to create a generic model for Latin medieval manuscripts. We used the recently released CATMuS-medieval model (Pinche 2023; Clérice et al. 2024; Pinche et al. 2024). Following the example of Lucas *et al.* (2022), we fine-tuned it

incrementally using two manuscripts as training data. We tested its performance on four manuscripts, two in-domain manuscripts (but with different page ranges) and two out-of-domain manuscripts.

Shelf-mark	Page range and lines	Date	Script	Layout
AM 242 fol.	4v–6r (training), 127 lines 8r-8v (testing), 64 lines	1340-1370	Gothic Textualis (influence from Cursive)	one column
AM 334 fol.	1v–3r (training), 296 lines 4r–4v (testing), 148 lines	1260-1281	Gothic Textualis	two columns
AM 132 fol.	130r-130v (testing), 167 lines	1330-1370	Gothic Textualis (influence from Cursive)	one column
AM 519 a 4to	2v-3r (testing) 70 lines	1270-1290	Gothic Textualis	two columns

Table 1 . An overview of the corpus.

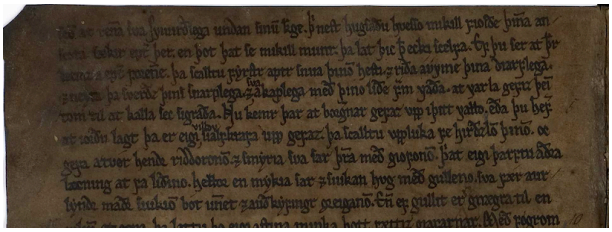

Figure 1 . *AM 519 a 4to, f. 2v, see full resolution images here: https://handrit.is/manuscript/view/da/AM04-0519-a/0?iabr=on#mode/2up*

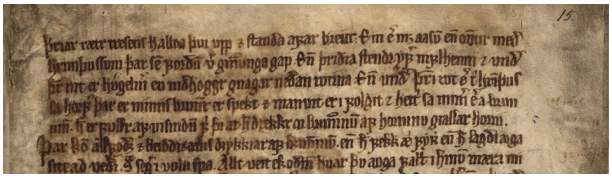

Figure 1 . *AM 519 a 4to, f. 2v, see full resolution images here: https://handrit.is/manuscript/view/da/AM04-0519-a/0?iabr=on#mode/2up* ***Figure 2 .*** *AM 242 fol., f. 8r, see full resolution images here: https://handrit.is/manuscript/view/da/AM02-0242/0?iabr=on#mode/2up*

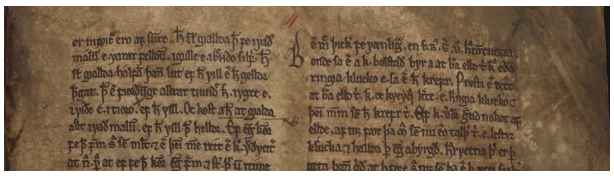

Figure 2 . *AM 242 fol., f. 8r, see full resolution images here: https://handrit.is/manuscript/view/da/AM02-0242/0?iabr=on#mode/2up* ***Figure 3 .*** *AM 334 fol., f. 4r, see full resolution images here: https://handrit.is/manuscript/view/is/AM02-0334/0?iabr=on#mode/2up*

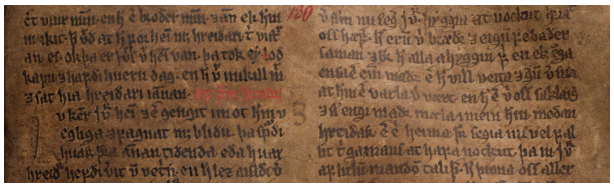

Figure 3 . *AM 334 fol., f. 4r, see full resolution images here: https://handrit.is/manuscript/view/is/AM02-0334/0?iabr=on#mode/2up* ***Figure 4 .*** *AM 132 fol., f. 130r, see full resolution images here: https://handrit.is/manuscript/view/is/AM02-0132/0?iabr=on#mode/2up*

Results and Discussion

The fine-tuning process of the CATMuS-medieval model demonstrates significant improvement in both character and word accuracy across in-domain and out-of-domain datasets. As presented in Table 2 and Figure 5, just eight pages of training data improved the model's performance by 20 percent points. Initially, at Step 0 (without fine-tuning), CATMuS-medieval exhibited character accuracy around 65.8% and word accuracy below 20%, with frequent errors in spacing, punctuation, similar letters ('i' vs. 'u'), and combining marks. Step 1 (two images in training) brought substantial gains, especially in-domain, with character accuracy rising to 83.1% and word accuracy surpassing 50%, while errors in spacing and punctuation decreased noticeably. Out-of-domain performance improved more slowly but steadily. By Step 2 (two more images), error rates for combining marks and rare characters decreased, with character accuracy reaching 85.2% in-domain and 72.8% out-of-domain. Word accuracy continued to rise, reaching 54.7% in-domain. Persistent issues included confusion between similar letters and combining marks. Step 3 marked further reduction in frequent errors, such as mistaking 'm' for 'n', with character accuracy climbing to 86.2% in-domain and 73.2% out-of-domain. Fi-

nally, Step 4 achieved the best results, with in-domain character accuracy at 88.0% and word accuracy at 63.0%. Out-of-domain improvements were slower but consistent.

Iteration	Number of pages used	Val Accuracy (Training)	In-domain character accuracy (Testing)	Out-of-domain character accuracy (Testing)
Step 0	0 (CATMuS)	-	0.658	0.689
Step 1	2 (1 per MS)	0.898	0.83	0.75
Step 2	4 (2 per MS)	0.885	0.852	0.749
Step 3	6 (3 per MS)	0.92	0.86	0.752
Step 4	8 (4 per MS)	0.913	0.88	0.782

Table 2 . Iterative training and testing.

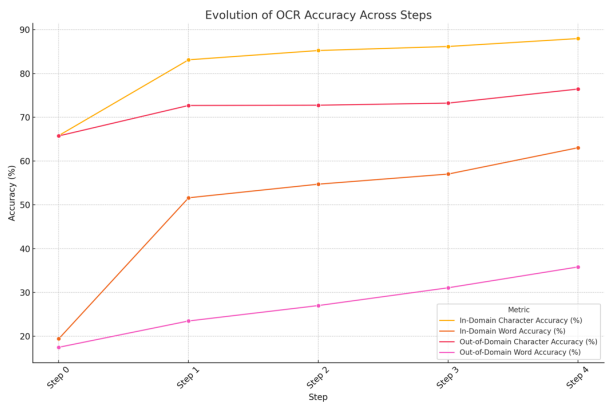

Figure 5. Evolution of accuracy (character and word accuracy) across steps. Step 0 represents the testing of CATMuS-medieval without fine-tuning.

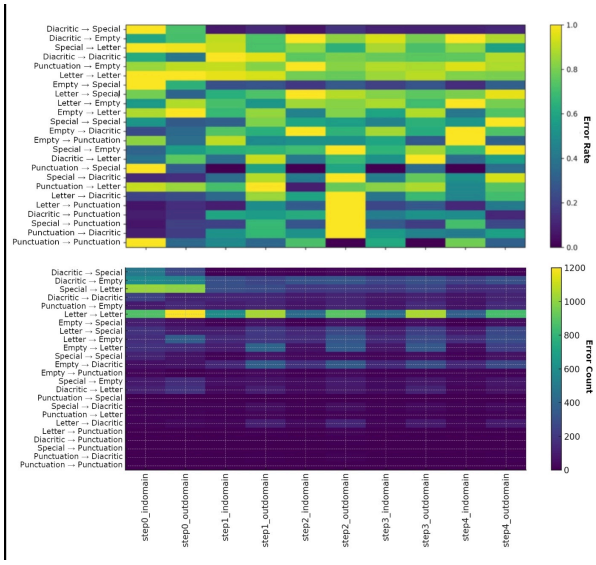

Figure 6. Heatmaps illustrating the evolution of the most frequent error types made by the model across steps, both in-domain and out-of-domain. Each row represents a specific confusion type with colour intensity indicating error rate (top) and error count (bottom). Brighter colours denote higher error frequencies.

Overall, as illustrated by the heatmaps in Figure 6, fine-tuning with incremental data led to effective error reduction in both in-domain and out-of-domain settings, with in-domain data benefiting more significantly. The improvement in the “Letter → Letter” error category (replacing one standard alphanumeric character with another) is the most pronounced, with absolute error counts decreasing across stages. Simultaneously, error rates also decrease substantially for punctuation, especially in the “Punctuation → Punctuation” and “Punctuation → Special” categories. Improvements across other categories, particularly those involving diacritics are less clear, suggesting that fine-tuning with a small amount of data primarily benefits standard character recognition rather than more complex or less frequent tokens.

Finally, as visually oriented humanists, we turned back to eScriptorium for a more visual representation of the performance. We managed to get some good lines, which were almost completely correct (Figure 7), but also some awful lines (Figure 8).

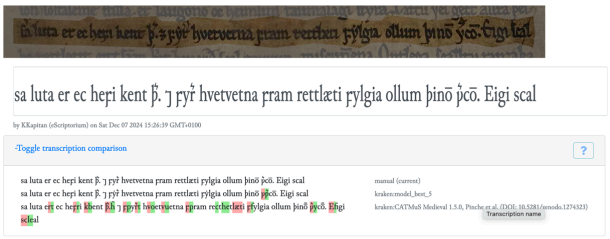

Figure 7. A success story: AM 519 a 4to, 2v:23. A screenshot showing the comparison of the manual transcription with the text predictions by CATMuS and our model in eScriptorium.

Figure 8. Not-so-much of a success story, AM 519 a 4to, 2v:7. A screenshot showing the comparison of the manual transcription with the text predictions by CATMuS and our model in eScriptorium.

The types of errors observed allow us to revise the transcription guidelines for the next iteration of our experiments. Originally, prioritising the characteristics of Old Norse-Icelandic manuscripts, we decided to add multiple special characters to the CATMuS character set. Among them were those representing the phonetic system of Old Norse-Icelandic, such as the vowels introduced by the First Grammatical Treatise ‘ǫ’, ‘ę’, ‘ø’, and the fricatives ‘þ’ and ‘ð’ (Guðvarður Már Gunnlaugsson 2008; Stefán Karlsson 2002). We also included some characters which appear in continental manuscripts but are ignored in CATMuS. For example, ‘r-rotunda’, which does not occur in the earliest Icelandic manuscripts but becomes increasingly common from the 13 th century onwards. Depending on which letters it follows, it gives us clues about chronology and may serve as evidence for the u-insertion, which appears around 1300 and becomes common by the late 14 th century. At the same time, to minimise the number of classes, we were not equally inclusive for all characters, e.g., ‘d’ and ‘uncial-d’, and this decision seems to confuse the model. When the manuscript features an ‘uncial-d’ for {d}, our model transcribes it as ‘ð’ (see Figure 8). This prompts further consideration of whether such characters should be reintroduced into the training data.

Therefore, these experiments allowed us to refine the principles and objectives for a larger-scale project whose primary goal is to expand the training dataset by producing transcriptions of new manuscripts. This will allow us to train new models in different configurations, test their performance, and assess the impact of an increased number of classes. Furthermore, it will facilitate discussions about the trade-offs between data modelling and the ability to address specific palaeographic questions regarding the chronological distribution of ‘r-rotunda’, ‘insular-f’, ‘uncial-d’, and the various shapes of the Tironian nota.

At the current stage of the project, our study demonstrates that even four pages per manuscript are sufficient for fine-tuning the CATMuS-medieval model, significantly improving its performance. Our results are promising, showcasing that HTR technology can be successfully used in Old Norse-Icelandic studies. Moreover, our case demonstrates that fine-tuning existing models with Kraken can be done easily and locally on an affordable machine, without requiring GPU access, or a subscription to external infrastructure, thus highlighting the accessibility of HTR technology today. We are finally able to offer a quick, open-access solution for scholars interested in incorporating HTR models into their research workflows, even if they are not directly involved in HTR development.

Notes

1. After our paper was accepted for the conference, we published the model and its underlying dataset on Zenodo (Kapitan and Vidal-Gorène 2025a, 2025b). Regardless of the publication date, it still represents the state of the project as of December 2024.

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